Authors' Rebuttal and List of Revisions

Joint MECC 2026 – ASME JDSMC Submission #26

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We thank the Associate Editor and the reviewers for their constructive feedback and valuable suggestions on our manuscript. We have carefully revised the manuscript to address all the points raised.

We are encouraged by the reviewers' recognition of the novelty and practical significance of our work. The Associate Editor described it as addressing *"an important and interesting research problem about modeling and parameter estimation of Diesel engine aftertreatment system with experimental validations."* Reviewer 14 noted that *"the proposed algorithm holds significant potential for practical engineering applications"* and that *"the paper is well-structured, tightly integrating theoretical derivation with engineering validation."* Reviewer 18 acknowledged the practical value of the approach for *"diagnosing catalyst aging in diesel Selective Catalytic Reduction (SCR) systems."*

Below we provide responses to each of the specific comments. Concerns shared by multiple reviewers are consolidated and addressed first.

Reviewers	**Reviewer Feedback**	**Rebuttal Comments and Revisions**
Associate Editor, Reviewer 18, Reviewer 14	*AE: "Justify the necessary condition of the proposed method based on sufficient data under vehicle working conditions."* *R14: "During real-world heavy-duty truck operations, the vehicle may encounter prolonged light-load cruising [. . .] which might result in entire driving segments failing to trigger catalyst saturation at all. Under such scenarios, will the system generate false alarms?"* *R18: "This is a severe bottleneck for real-world deployment, evidenced by the fact that the actual truck drive segments [. . .] featured very few samples (often under 10%) near saturation."*	Although only a fraction of the operating data is close to saturation (approximately 10% in real-world truck data), the convex parameter estimation framework utilizes the entire dataset. The non-saturated data points establish the lower boundary constraints ($\eta(k) \leq \eta_{\text{sat}}(k)$), which restrict the parameter space and prevent the estimates from collapsing. For on-vehicle implementation, catalyst utilization can be assessed in real time by monitoring readily available signals such as urea dosing rate, engine rpm, and the NO_x reduction estimate from sensors. When these indicate sustained low catalyst demand, the system defers its decision rather than executing the hypothesis test on insufficient data. The detector can thus avoid producing false alarms under non-catalyst-saturating driving conditions. **Revision:** A dedicated paragraph discussing these practical OBD considerations, false-alarm prevention, and the impact on the in-use monitor performance ratio (IUPMR) has been added at the end of Section 6. A note on monitoring catalyst utilization via available signals has been added to the Limitations appendix (Appendix B).

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Reviewer 14	*"In the test-cell validation, the decision thresholds are set to $\beta_{hFTP} = 4.00$ and $\beta_{RMC} = 10.00$, respectively. Why is the threshold abruptly scaled up directly to $\beta_{truck} = 250$ in the real-world truck driving segment analysis?"*	The threshold $\beta_{\text{truck}} = 250$ is significantly larger than the test-cell thresholds because the limited sample size causes T_w to diverge from its asymptotic χ^2_ν distribution under $\mathcal{H}_0$. Truck data has considerably fewer saturated-mode samples ($N_{\text{sat}} \approx 120$–$150$) compared to test-cell experiments ($N_{\text{sat}} \approx 200$–$500$), resulting in heavier tails in the null distribution. This is compounded by the less reliable estimation of the error variance under road conditions and the unaccounted variance of the reference parameter θ^{dg}_{sat}. **Revision:** Two paragraphs have been added at the end of Section 6 justifying the threshold discrepancy and discussing the path toward statistically optimal threshold selection.
Reviewer 18	*"The saturated catalyst model's response does not depend on the previous state but only on the inputs, making the response independent of the initial conditions. This effectively means [. . .] that the dynamics are irrelevant, and that is the reason a simple MLE point estimator suffices. In that sense, this paper does not fit the scope of MECC, because the system dynamics don't really matter."*	The goal of the referenced phrase was to convey that one mode (the saturated mode) of the switched dynamic model for NO_x reduction is independent of initial conditions. This property allows us to use the prediction error to classify data points as "close to saturation" when computing the statistics for aging detection. The next time-step NO_x reduction still depends on current inputs through a discrete-time equation. The saturated-mode model is not a static relationship—it is a discrete-time input-output model derived from molar conservation laws. The MLE estimator exploits the time-series structure of the saturated-mode dynamics. The non-saturated data points contribute dynamic inequality constraints. The overall framework is fundamentally a dynamic systems identification and change detection problem. The requirement of catalyst saturation is a deliberate design choice that enables the core novelty: saturated operation simplifies the highly nonlinear, unobservable ammonia coverage dynamics into a static limit, allowing a convex parameter estimation framework without requiring an on-board ammonia sensor. **Revision:** Additional clarification has been added to the first paragraph of Section 4 to better convey the role of mode independence within the switched dynamic framework.
Reviewer 12	*"The significance of this paper should be stated clearly."*	We have added a dedicated paragraph in the Introduction explicitly outlining the three main contributions: (1) a first-principles discrete-time input-output model derived from molar conservation that captures the interplay between gas parcel residence time and sensor sampling intervals; (2) a convex parameter estimation framework that bypasses the need for on-board NH_3 sensors while remaining robust to the unidirectional bias of commercial NO_x sensors; and (3) a Wald-test-based catalyst aging detector validated using both test-cell and real-world truck driving datasets. **Revision:** The significance paragraph has been added to the end of Section 1 (Introduction).

Reviewers	Reviewer Feedback	Rebuttal Comments and Revisions
Reviewer 12	*"The readers may want to see the clear comparison of experimental data and model results. It is better to present the figures but the numbers."*	The detection results span multiple trucks, multiple drive segments, and two data years. Presenting all results as individual time-series plots would obscure the cross-comparison that the tabular format enables. The existing tables (Tables 5–9) summarize the detection results for all trucks and drive segments in a compact format that facilitates direct comparison of aging detection outcomes. **Revision:** No additional figures were added. The tabular format was retained as the most effective presentation for the scope of the results.
Reviewer 12	*"The literature review is not sufficient. The limitations of the existing works are not clear."*	The paper's references are organized around three pillars: SCR modeling and control, SCR aging diagnostics, and the underlying mathematical and statistical theory. These references collectively justify the research objective, identify the gap, and map directly to the specific focus of model-based SCR aging diagnostics under catalyst saturation. **Revision:** Additional references to justify the β discrepancy were added.
Associate Editor, Reviewer 12, Reviewer 14	*AE: "Some writing issues (e.g., Appendix-?? on page 5) need to be revised."* *R12: "There are some writing mistakes in this paper. Such as: the expression of '(Appendix-??)' is wrong."* *R14: "NO_x, NH_3, O_2, and N_2 must be strictly rendered in roman font."*	1. The writing issues were due to a missing cross-reference link caused by a LaTeX label mismatch while converting the paper from the JDSMC template to the MECC template. The broken cross-reference has been corrected and now compiles as (Appendix B) in Section 3. 2. All chemical symbols have been updated to roman font (NO_x, NH_3, O_2, N_2) throughout the text, as suggested by Reviewer 14. **Revision:** Cross-reference fixed in Section 3; chemical symbols corrected throughout.